

Common Lyapunov Function for Exponentially Stable Nonlinear Systems

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Abstract

We present a sufficient condition for the existence of a common Lyapunov function for a family of exponentially stable nonlinear systems. Suppose that there are m systems $\dot{x} = f_i(x)$ each of which is exponentially stable. When any pair of the vector fields $f_i(x)$ are commuting, i.e. the Lie bracket of the pair is zero, there is one Lyapunov function $V(x)$ which guarantees the exponential stability for all of them.

I. INTRODUCTION

Converse Lyapunov theorem, which deals with the existence of a Lyapunov function for a stable system, has been well studied in 1950-60. These results have been primarily of theoretical significance, and the problem is satisfactorily answered. However, the existence of a common Lyapunov function for several stable systems, i.e., whether there is one Lyapunov function $V(x)$ for all stable systems

$$\dot{x} = f_1(x) \quad \dot{x} = f_2(x) \quad \cdots \quad \dot{x} = f_m(x),$$

are still being studied. Recent growth of intelligent control or switching control scheme calls for this type of Lyapunov function for guaranteeing their stability, and the utility of the existence of such Lyapunov function is a kind of theoretical applications.

A simple and intuitive answer for this problem is given in [1]. They assumed the system is linear, and suggested a sufficient condition that all system matrix A_i 's are commuting. This work is also extended for a class of linear systems in [2]. However, the key point of the work [1] is commutability of the trajectories of the systems, which is naturally applicable to nonlinear systems.

In this paper, we generalize the work of [1] inspired from [3]. For the convenience, some notational definitions are given here. $\hat{x}_i(t, x)$ is the solution of $\dot{x} = f_i(x)$ starting at x when $t = 0$. Also, another notation $\phi_i(t, x, t_0)$ denotes the solution determined by the initial time t_0 and the initial point x , so that $\phi_i(t_0, x, t_0) = x$. A function is said to be C^k if it is continuously differentiable k times. $|\cdot|$ stands for the Euclidean norm of a vector. B_r stands for a compact ball of radius r from the origin.

II. COMMON LYAPUNOV FUNCTION FOR COMMUTING VECTOR FIELDS

Suppose a family of nonlinear systems

$$\dot{x} = f_i(x), \quad i = 1, 2, \dots, m. \quad (1)$$

where f_i is C^k ($k \geq 1$) function and $f_i(0) = 0$.

Assume that all the systems are exponentially stable. More specifically, there exist constants $k, \lambda, r > 0$ such that

$$|\tilde{x}_i(t, x)| \leq k |x| e^{-\lambda t}, \quad \forall t \geq 0, \quad \forall x \in B_r \quad (2)$$

where $i = 1, 2, \dots, m$. The existence of such k and λ for all the systems has no loss of generality since, from the exponential stability of each system, k can be chosen to be maximal, and λ to be minimal among k_i 's and λ_i 's of each systems.

Since f_i 's are C^k , there exists a constant L such that

$$\left| \frac{\partial f_i}{\partial x}(x) \right| \leq L, \quad \forall x \in B_{kr} \quad (3)$$

which is needed in the forthcoming argument.

Before the main result, a well-known fact is repeated here for convenience. (See e.g. [8, p.51].)

Lemma 1. *For the nonlinear systems (1), the trajectories of any two systems are commuting, i.e.,*

$$\tilde{x}_i(t, \tilde{x}_j(s, x)) = \tilde{x}_j(s, \tilde{x}_i(t, x)), \quad \text{for all } t, s \geq 0 \quad (4)$$

if and only if

$$[f_i, f_j](x) = 0 \quad (5)$$

where $1 \leq i, j \leq m$ and $x \in B_{kr}$.

Now, we construct a common Lyapunov function for all the systems by the following iterative procedure.

$$\begin{aligned} V_1(x) &= \int_0^T |\tilde{x}_1(s, x)|^2 ds \\ V_i(x) &= \int_0^T V_{i-1}(\tilde{x}_i(s, x)) ds, \quad i = 2, \dots, m. \end{aligned} \quad (6)$$

where T is a finite constant to be chosen later. In the followings, we show that the $V_m(x)$ is a common Lyapunov function for the commuting and exponentially stable systems (1).

Lemma 2. *Let a sequence $\{\sigma(1), \sigma(2), \dots, \sigma(i)\}$ is a permuted sequence of $\{1, 2, \dots, i\}$. Under the assumption that $[f_j, f_k](x) = 0$ for $1 \leq j, k \leq i$, the function V_i can be constructed by any permuted sequence of $\{1, 2, \dots, i\}$, that is,*

$$V_i(x) = \int_0^T \cdots \int_0^T \int_0^T |\tilde{x}_1(s_1, \tilde{x}_2(s_2, \dots, \tilde{x}_i(s_i, x) \cdots))|^2 ds_1 ds_2 \cdots ds_i \quad (7)$$

$$= \int_0^T \cdots \int_0^T \int_0^T |\tilde{x}_{\sigma(1)}(s_{\sigma(1)}, \tilde{x}_{\sigma(2)}(s_{\sigma(2)}, \dots, \tilde{x}_{\sigma(i)}(s_{\sigma(i)}, x) \cdots))|^2 ds_{\sigma(1)} ds_{\sigma(2)} \cdots ds_{\sigma(i)}. \quad (8)$$

Proof. Suppose that $i = 2$. Then, by Lemma 1,

$$\begin{aligned} V_2(x) &= \int_0^T \int_0^T |\tilde{x}_1(s_1, \tilde{x}_2(s_2, x))|^2 ds_1 ds_2 \\ &= \int_0^T \int_0^T |\tilde{x}_1(s_1, \tilde{x}_2(s_2, x))|^2 ds_2 ds_1 \\ &= \int_0^T \int_0^T |\tilde{x}_2(s_2, \tilde{x}_1(s_1, x))|^2 ds_2 ds_1 \end{aligned}$$

Hence, the lemma holds for any permuted sequence of $\{1, 2\}$, and the extension to $i > 2$ is trivial. $\square$

Lemma 3. *Assume that all the systems of (1) are exponentially stable as in (2), and that all vector fields are commuting as in (5). Then, for $i = 1, \dots, m$, the function $V_i(x) : B_{r/k^{i-1}}(\subset R^n) \rightarrow R$ as defined in (6) satisfies the followings*

$$c_1^i |x|^2 \leq V_i(x) \leq c_2^i |x|^2 \quad (9)$$

$$L_{f_j} V_i \leq -\beta_i |x|^2, \quad j = 1, \dots, i. \quad (10)$$

$$\left| \frac{\partial V_i}{\partial x} \right| \leq \frac{1}{2} c_3^i |x| \quad (11)$$

where

$$c_1 = \frac{1}{2L}(1 - e^{-2LT}) \quad (12)$$

$$c_2 = \frac{k^2}{2\lambda}(1 - e^{-2\lambda T}) \quad (13)$$

$$c_3 = \frac{k}{\lambda - L}(1 - e^{-(\lambda - L)T}) \quad (14)$$

$$\beta_i = c_1^{i-1} - c_2^{i-1} k^2 e^{-2\lambda T} \quad (15)$$

and these constants are all positive with an appropriate T .

Proof. We prove the lemma by induction. First, suppose V_i satisfies the lemma, especially (9) and (11). Then, for

$$x \in B_{r/k^i},$$

$$\begin{aligned} V_{i+1}(x) &= \int_0^T V_i(\tilde{x}_{i+1}(s, x)) \, ds \\ &\leq \int_0^T c_2^i |\tilde{x}_{i+1}(s, x)|^2 \, ds \\ &\leq c_2^i k^2 |x|^2 \int_0^T e^{-2\lambda s} \, ds \\ &= c_2^i \frac{k^2}{2\lambda} (1 - e^{-2\lambda T}) |x|^2 \\ &= c_2^{i+1} |x|^2 \end{aligned}$$

by (2) and (13). And, from (3), it can be easily shown that [5, p.93]

$$|\tilde{x}_{i+1}(s, x)| \geq |x| e^{-Ls}. \quad (16)$$

Hence,

$$\begin{aligned} V_{i+1}(x) &= \int_0^T V_i(\tilde{x}_{i+1}(s, x)) \, ds \\ &\geq \int_0^T c_1^i |\tilde{x}_{i+1}(s, x)|^2 \, ds \\ &\geq c_1^i |x|^2 \int_0^T e^{-2Ls} \, ds \\ &= c_1^i \frac{1}{2L} (1 - e^{-2LT}) |x|^2 \\ &= c_1^{i+1} |x|^2. \end{aligned}$$

Another easy calculation shows that [4, p.241]

$$\left| \frac{\partial \tilde{x}_{i+1}}{\partial x}(s, x) \right| \leq e^{Ls}. \quad (17)$$

Therefore,

$$\begin{aligned} \left| \frac{\partial V_{i+1}}{\partial x} \right| &\leq \int_0^T \left| \frac{\partial V_i}{\partial x}(\tilde{x}_{i+1}(s, x)) \right| \left| \frac{\partial \tilde{x}_{i+1}}{\partial x} \right| \, ds \\ &\leq \int_0^T \frac{1}{2} c_3^i |\tilde{x}_{i+1}(s, x)| e^{Ls} \, ds \\ &\leq \frac{1}{2} c_3^i k |x| \int_0^T e^{-\lambda s} e^{Ls} \, ds \\ &= \frac{1}{2} c_3^i \frac{k}{\lambda - L} (1 - e^{-(\lambda - L)T}) |x| \\ &= \frac{1}{2} c_3^{i+1} |x|. \end{aligned}$$

Now, before evaluating $L_{f_j} V_{i+1}$, consider V_{i+1} as

$$\begin{aligned} V_{i+1}(x(t)) &= \int_0^T V_i(\tilde{x}_{i+1}(s, x(t))) \, ds \\ &= \int_t^{t+T} V_i(\phi_{i+1}(\tau, x(t), t)) \, d\tau. \end{aligned}$$

In the above equation, $\phi_{i+1}(\tau, x(t), t)$ is independent of t along the trajectory of f_{i+1} [7, p.232]. Indeed,

$$\begin{aligned} \frac{\partial \phi_{i+1}(\tau, x(t), t)}{\partial t} &= \lim_{h \rightarrow 0} \frac{\phi_{i+1}(\tau, \phi_{i+1}(t+h, x(t), t), t+h) - \phi_{i+1}(\tau, \phi_{i+1}(t, x(t), t), t)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\phi_{i+1}(\tau, x(t), t) - \phi_{i+1}(\tau, x(t), t)}{h} \\ &= 0. \end{aligned}$$

Evaluating $L_{f_{i+1}} V_{i+1}$ which is the derivative of V_{i+1} along the trajectory of f_{i+1} ,

$$\begin{aligned} L_{f_{i+1}} V_{i+1} &= \dot{V}_{i+1}(x(t)) = V_i(\phi_{i+1}(t+T, x(t), t)) - V_i(\phi_{i+1}(t, x(t), t)) \\ &= V_i(\tilde{x}_{i+1}(T, x)) - V_i(x) \\ &\leq c_2^i |\tilde{x}_{i+1}(T, x)|^2 - c_1^i |x|^2 \\ &\leq c_2^i k^2 e^{-2\lambda T} |x|^2 - c_1^i |x|^2 \\ &= -(c_1^i - c_2^i k^2 e^{-2\lambda T}) |x|^2. \end{aligned}$$

On the other hand, for $j = 1, \dots, i$, the function $L_{f_j} V_{i+1}$ can be evaluated on the same bases. Indeed, by Lemma 2, there exists a function $W(x)$, which satisfies $c_1^i |x|^2 \leq W(x) \leq c_2^i |x|^2$, such that,

$$V_{i+1}(x) = \int_0^T W(\tilde{x}_j(s, x)) ds.$$

By the same reasoning as $L_{f_{i+1}} V_{i+1}$, it can be easily shown that

$$L_{f_j} V_{i+1} \leq -(c_1^i - c_2^i k^2 e^{-2\lambda T}) |x|^2.$$

This proves the inequality of (10).

Now, it is left to show that the lemma holds for $i = 1$. However, this is an easy application of the standard converse theorem [4-7]. (Especially refer to [5, p.180].) Also, it can be shown directly according to the above arguments. So, it is omitted here.

Finally, there exists a T^* such that, for any $T \geq T^*$,

$$\left(\frac{c_1}{c_2}\right)^{i-1} = \left[\frac{\lambda}{Lk^2} \frac{(1 - e^{-2LT})}{(1 - e^{-2\lambda T})}\right]^{i-1} > k^2 e^{-2\lambda T}, \quad \forall i \in [1, m] \quad (18)$$

since the left hand converges to $\frac{\lambda}{Lk^2}$ and the right hand converges to zero as $T \rightarrow \infty$. Choosing any $T > T^*$, it can be shown all the constants (12) - (15) are positive. Indeed, by (18), for any $1 \leq i \leq m$,

$$\beta_i = c_1^{i-1} - c_2^{i-1} k^2 e^{-2\lambda T} > 0,$$

and c_1 , c_2 and c_3 are readily positive by their forms. This completes the proof. $\square$

Now we can claim the following Theorem.

Theorem 4. Consider a family of nonlinear systems (1). When the systems are exponentially stable in the sense of (2) and commuting on B_{kr} in the sense of (5), the family has one common Lyapunov function $V(x)$ on $B_{r/k^{m-1}}$ satisfying, with positive constants a_1 , a_2 , a_3 and a_4 ,

$$a_1 |x|^2 \leq V(x) \leq a_2 |x|^2 \quad (19)$$

$$\left| \frac{\partial V}{\partial x} \right| \leq a_3 |x| \quad (20)$$

$$L_{f_j} V \leq -a_4 |x|^2, \quad \forall j \in [1, m] \quad (21)$$

Moreover, one constructing method for such Lyapunov function is as in (6).

The proof follows from Lemma 1, 2 and 3 by taking $V(x) = V_m(x)$.

Remark 1. The commuting property of f_i 's is needed only for (10). The property (9) and (11) is preserved by iterative integration along a trajectory of any exponentially stable system.

Remark 2. In the definition of Lyapunov function (6), a finite T is used. A finite T guarantees that c_3 is positive constant in (14). However, in the Lyapunov function with $T = \infty$, other properties aren't broken, but only the inequality (11) is not guaranteed.

Remark 3. When the inequality (3) holds globally, then a common global Lyapunov function also can be constructed instead of on the region $B_{r/k^{m-1}}$.

III. CONCLUSION

In this paper, it is shown that when a family of nonlinear systems are exponentially stable and commuting, then there is a common Lyapunov function for the family. This results also shows that the crucial point of [1] is just interchangeability of the trajectories of vector fields $A_i x$ and thus the results of [1] is applicable to nonlinear cases.

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